Propositions

Definition (proposition): A proposition, or statement, is a declarative sentence which is either true or false

For example, "the Sun is white", "$1+2=3$", and "$1+2=4$" are all propositions.
However, "42" and "$1+2$" are not propositions.

Definition (connectives): Connectives, or logical operators, or truth functions, are defined by means of truth tables. The symbols p and q denote propositions; T stands for "true" and F stands for "false".

$\wedge$ and
$\vee$ or
$\neg$ not
$\Rightarrow$ implies
$\Leftarrow$ implied by
$\Leftrightarrow$ if and only if (iff)

Here are some truth table examples:

p	$\neg p$
F	T
T	F

p	q	$p \wedge q$	$p \vee q$	$p \Rightarrow q$
F	F	F	F	T
F	T	F	T	T
T	F	F	T	F
T	T	T	T	T

All logical possibilities in table

If two propositions are logically equivalent, then it will be reflected in the truth table. For instance:

p	q	$p \wedge \neg q$	$\neg(p \Rightarrow q)$
F	F	F	F
F	T	F	F
T	F	T	T
T	T	F	F

Therefore, $(p \wedge \neg q)$ is logically equivalent to $\neg(p \Rightarrow q)$

Definition (Tautologies and contradictions): A sentence that is always true is called a tautology, whilst a statement that is always false is called a contradiction.

Here are some useful tautologies:

$p \Rightarrow (p \vee q)$ is known as addition

$(p \wedge q) \Rightarrow p$ is known as simplification

$[p \wedge (p \Rightarrow q)] \Rightarrow q$ (Modus Ponens)

$(\neg p \Rightarrow c) \Rightarrow p$ and $(p \Rightarrow q) \Leftrightarrow [(p \wedge \neg q) \Rightarrow c]$ (Reductio ad Absurdum)

$\neg(p \wedge q) \Leftrightarrow (\neg p \vee \neg q)$ (De Morgan's law)

$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$ (Contrapositive law)

Sets

A **set** is a collection of objects known as elements. For example, $\{1, 2, 3\}$ is the set of numbers 1, 2, and 3.

Some well known sets are:

- $\mathbb{N}$: the set of natural numbers $\{1, 2, 3, \ldots\}$
- $\mathbb{Z}$: the set of integers $\{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\}$
- $\mathbb{Q}$: the set of rational numbers $\{\frac{m}{n} : (m, n \in \mathbb{Z}) \wedge (n \neq 0)\}$

$\in$ simply means in

- $\mathbb{R}$: the set of real numbers (can be written using decimals)
- $\emptyset$: the empty set

$\in$ means "in" or "is an element of", so $a \in A$ means a is an element of A.

Definition (Universal and existential quantifiers): $\forall$ means "for all" or "for each", whilst $\exists$ means "there exists" or "there is"

To negate a proposition containing quantifiers, flip the quantifier and truth:

$$\neg[(\forall x \in A)(P(x))] \rightarrow (\exists x \in A)(P(x))$$
or
$$(\forall x \in A)(\neg P(x))$$

Definition (subset): Let A and B be sets. A is a subset of B if for all $x \in A, x \in B$. This is denoted as $A \subseteq B$.

Sets can be built from other sets or properties:

$$S = \{x \in A \mid P(x)\}$$

For instance, $\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{Z} \wedge n \neq 0\}$

Definition (Set operations): Let X be a universe of discourse and let A and B be subsets of X.

Then:

- The intersection of A and B is denoted using $A \cap B = \{x \in X : x \in A \wedge x \in B\}$
- The union of A and B is denoted using $A \cup B = \{x \in X : x \in A \vee x \in B\}$
- The relative complement of B in A is denoted with $A - B$ or $A \setminus B = \{x \in X : x \in A \wedge x \notin B\}$

Definition (convection): Let X be an acknowledged universe of discourse, and let A be a subset of X. The relative component of A in X is denoted A^c, known as the complement of A.

Definition (Set equality): Let A and B be sets. The set A and B are equal if $A \subseteq B$ and $B \subseteq A$.